Validation-Gated Decision-Level Evidence for Cross-Subject TMS-EEG Depression Screening

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***Abstract*— Cross-subject depression screening from transcranial magnetic stimulation electroencephalography (TMS-EEG) remains difficult because subject-specific variability can dominate disease-related neural responses. We study where auxiliary evidence should enter a strictly subject-independent waveform pipeline. On a fixed 18-subject test split, a decision-level Riemannian residual changes subject-level F1 from 0.6625 to 0.7662 by correcting two false negatives without adding false positives; however, specificity remains 0.5556 and the paired bootstrap interval crosses zero. Five-fold nested evaluation over all 89 subjects does not confirm a stable residual advantage. Validation-gated non-diagnostic metadata changes nested subject-level F1 from 0.5778±0.0617 to 0.6384±0.0850, with improvement in three of five folds. The evidence supports EA-Clean-STGNN as the default waveform model; auxiliary evidence remains exploratory and is limited to research-oriented screening.**

***Index Terms*—TMS-EEG, depression screening, cross-subject generalization, Riemannian geometry, graph neural networks.**

I. INTRODUCTION

TMS-EEG provides a noninvasive way to probe cortical excitability and network-level response dynamics, making it attractive for objective depression assessment [1]–[3]. EEG-based depression screening and recent TMS-EEG studies further motivate carefully validated subject-level models [4]–[7]. The main obstacle is cross-subject generalization. EEG responses vary with individual anatomy, electrode placement, scalp impedance, trial quality, and state-dependent neural fluctuations. Consequently, a model that performs well on source-subject epochs can fail on unseen subjects.

This is particularly important for clinical screening, where the final endpoint is a subject-level decision rather than isolated epoch predictions. The model is intended to support triage and prioritization for follow-up assessment; a structured clinical interview remains necessary. A rigorous protocol must therefore separate training, validation, and held-out test subjects, and it must avoid selecting checkpoints or thresholds using test behavior.

Cross-subject EEG methods commonly rely on alignment, transfer, or deep representation learning [8], [9]. EA removes broad covariance-scale differences while preserving the waveform tensor expected by temporal and graph neural networks [10]. Riemannian methods are theoretically well matched to EEG covariance matrices because spatial covariance matrices lie on the symmetric positive definite manifold [11], [12]. Feature adaptation methods attempt to normalize hidden representations using subject-conditioned statistics. In our TMS-EEG depression setting, however, increasingly invasive feature-level adaptation proved fragile.

TMS-EEG studies have examined evoked-potential markers of MDD severity and cognition, while EEG studies emphasize the need for independent validation of predictive models [5], [6], [13]. BENDR, EEG Conformer, BIOT, and LaBraM use convolutional Transformers, cross-dataset learning, or self-supervised pretraining to learn transferable EEG representations [14]–[17]. Comparisons under the same subject-exclusive protocol would clarify how these encoders perform relative to decision-level evidence integration.

This paper therefore studies a conservative question: can auxiliary evidence help without perturbing the waveform-graph representation? We retain a clean EA-preprocessed STGNN backbone and first add Riemannian tangent-space evidence only at the decision level:

$$z_{\text{final}} = z_{\text{clean}} + \lambda z_{\text{RA}}, \tag{1}$$

where z_{clean} is the clean STGNN logit, z_{RA} is an auxiliary Riemannian logit, and λ is selected only on source-validation subjects. We then test whether non-diagnostic metadata can provide a separate non-waveform auxiliary source under validation-gated activation.

Our contributions are three-fold. First, we establish a subject-exclusive TMS-EEG evaluation in which the subject, rather than the epoch, is the independent clinical unit. Second, controlled negative results show that direct Riemannian preprocessing and subject-conditioned feature adaptation are fragile under the current waveform STGNN representation; the observed failure modes are specific to the present task and evaluation protocol. Third, we compare decision-level Riemannian, metadata, and validation-gated auxiliary evidence while separating fixed-split mechanism analysis from nested internal robustness.

II. METHODS

A. Dataset Representation and Protocol

The dataset contains 8,307 TMS-EEG epochs from 89 subjects (44 controls and 45 depression subjects). Each epoch is represented as a $10 \times 62 \times 50$ tensor: 62 channels, 500 samples, and 10 temporal patches. The task is binary control/depression screening. Because epochs from the same person are correlated, the subject is the independent evaluation unit; epoch F1 is reported only as a secondary diagnostic. The main endpoint is weighted subject-level F1, accompanied by

TABLE I
SUBJECT-INDEPENDENT PROTOCOL SUMMARY. LABEL COUNTS ARE CONTROL/DEPRESSION.

Stage	Split	N_s	N_e	C/D
Strict validation	Train	55	5153	2532/2621
Strict validation	Validation	16	1459	721/738
Strict validation	Held-out test	18	1695	847/848
Final evaluation	Source calibration	71	6612	3253/3359
Final evaluation	Held-out test	18	1695	847/848

TABLE II
COHORT CHARACTERISTICS. HAMD IS SHOWN ONLY FOR COHORT DESCRIPTION AND IS EXCLUDED FROM ALL MODEL INPUTS.

Group	N_s	Age	Female	MT	HAMD
Control	44	46.1±13.1	25 (56.8%)	0.6±0.1	1.2±1.8
Depression	45	43.6±13.2	26 (57.8%)	0.6±0.1	28.5±8.7
All	89	44.9±13.1	51 (57.3%)	0.6±0.1	15.0±15.0

sensitivity, specificity, balanced accuracy, AUC, and confusion matrix. Table II reports cohort-level clinical and stimulation descriptors. HAMD is included only for cohort characterization and is never used as a model input.

Table I summarizes the evaluation protocol. Hyperparameters are first selected using source-subject validation. After selection, the final configuration is recalibrated on all 71 source subjects and evaluated once on the 18 held-out test subjects (9 controls and 9 depression subjects). This fixed split is used to inspect how auxiliary evidence changes decisions. Because a single 18-subject test has wide uncertainty, we additionally perform five-fold nested subject-level evaluation over all 89 subjects to assess internal robustness across partitions. Both analyses are internal evaluations of the same cohort.

Table II reports only de-identified aggregate characteristics. Recruitment-site identifiers and record-level clinical information are withheld from the blinded manuscript and public repository to preserve double-blind review and participant privacy. After verification against the governed protocol, ethics documentation, and clinical records, a de-identified supplementary provenance table will report the recruitment window at a non-identifying resolution, diagnostic and eligibility procedures, treatment categories, variable completeness, and confounding handling. Institution- and approval-specific ethics and consent documentation will be provided confidentially to the editor or restored in the unblinded final manuscript, as required by journal policy.

B. EA-Clean-STGNN Backbone

The clean backbone applies EA to each subject's waveform epochs and then trains an STGNN without subject-conditioned normalization, N-ASN, PSNA, or feature-level adaptation. EA reduces broad subject-specific covariance differences while preserving the original waveform tensor. The STGNN combines temporal convolution with graph message passing over EEG channels, capturing both TMS-evoked waveform dynamics and spatial channel interactions. Compact EEG backbones such as EEGNet [18] and static or dynamic graph neural models [19], [20] provide the broader architectural context for this design. The spatial graph is constructed dynamically rather than fixed from a template montage. For each sliding spatio-temporal window, node features are used to form a dot-product similarity matrix, followed by LeakyReLU, row-wise softmax normalization, and an added self-loop identity term. Thus, the adjacency matrix is sample- and window-adaptive, while the message-passing parameters are optimized end-to-end.

Algorithm 1 Validation-Gated Decision-Level Auxiliary Evidence

Require: TMS-EEG epochs X_s and non-diagnostic metadata m_s for subject s; trained STGNN; trained metadata classifier f_{meta}; validation-selected λ, gating policy g_s, and threshold θ.

Apply EA to waveform epochs.
Compute clean logits z_{clean} for each epoch.
Estimate stabilized covariance and tangent-space features.
Compute auxiliary Riemannian logits z_{RA}.
Fuse Riemannian evidence by Eq. (1) when RA residual inference is used.
Compute metadata probability $p_{\text{meta}} = f_{\text{meta}}(m_s)$.
Convert clean or residual epoch logits to probabilities and aggregate by subject mean.
Apply metadata only when $g_s = 1$ and its confidence passes θ; otherwise keep the EEG-only subject probability.
Optionally reject high-entropy epochs using a validation-selected threshold.
Return subject-level probability and class prediction.

C. Decision-Level Riemannian Residual

The Riemannian branch is deliberately decoupled from the STGNN feature extractor. For each epoch, the $10 \times 62 \times 50$ tensor is reshaped into a full-window 62×500 channel-time matrix. A trace-stabilized covariance matrix is computed, mapped to a log-eigen tangent representation, vectorized from the upper triangular entries, and passed through a small classifier to produce z_{RA}. The selected configuration uses Riemannian shrinkage $\alpha = 0.2$, fractional whitening $\rho = 0.5$, calibration epoch 7, and $\lambda = 0.10$; $\lambda = 0.08$ is retained as a robustness check. To ensure scale compatibility between the two branches, both z_{clean} and z_{RA} are raw, unnormalized logits output by their final classifiers. The residual coefficient λ therefore acts as an implicit scale-matching and decision trade-off parameter and is selected strictly through validation sweeps.

Algorithm 1 gives the complete inference path. The central design rule is that Riemannian evidence may shift the final decision but never normalizes or rewrites STGNN hidden features.

D. Non-Diagnostic Metadata Branch

To test whether a separate non-waveform evidence source can improve decision behavior, we evaluate a subject-level

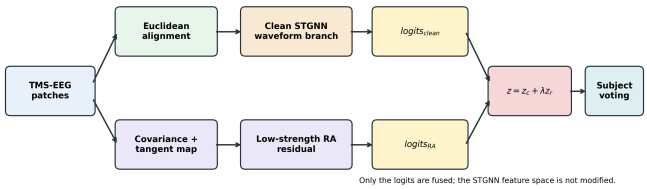


Fig. 1. Conservative decision-level fusion. The EA-Clean-STGNN branch preserves waveform features, while Riemannian evidence contributes only a small logit-level residual.

metadata branch, following the broader motivation for late fusion of clinical context [21]. To avoid label or protocol leakage, we exclude HAMD, treatment arm, randomization group, treatment response, and subject identifier. Only non-diagnostic baseline variables are used: age, sex, motor threshold, and stimulation intensity. These variables do not encode clinician-rated depression severity, treatment assignment, treatment response, or any post-hoc diagnostic outcome. The metadata table covers all 89 subjects and is matched to the TMS-EEG subject IDs using the same preprocessing order as the subject-ID dataset. Label consistency is checked in every fold, but labels from the held-out fold are never used to select metadata features, thresholds, or gating policies.

Within each outer fold, a low-capacity logistic-regression metadata classifier is fit only on source-training subjects. Feature subset, regularization, fusion strength, and threshold are selected only on source-validation subjects.

E. Validation-Gated Auxiliary Evidence

The validation-gated rule keeps Clean STGNN as the default predictor and activates metadata evidence only when a validation-selected policy indicates that the auxiliary branch is reliable. Candidate policies include specificity guards, sensitivity guards, and bidirectional guards based on clean–metadata disagreement and metadata confidence. All policy type, confidence threshold, feature subset, and classifier regularization parameters are selected only on source-validation subjects. The selected policy is then applied unchanged to the outer test subjects.

This design applies auxiliary evidence selectively. If no validation-selected guard is activated for a test subject, the final decision is exactly the Clean STGNN decision. This prevents the metadata branch from overwriting a stable EEG prediction when validation behavior does not support doing so.

F. Implementation Details

Table III summarizes the main implementation settings used for the Clean STGNN and auxiliary branches. Checkpoints are selected on source-validation subjects before held-out evaluation. The held-out test split is not used for model selection, threshold selection, or entropy-gating selection.

The Clean model contains 26,495,204 trainable parameters and is therefore large relative to 89 independent subjects. Subject-exclusive splits, source-validation selection, dropout, weight decay, nested evaluation, and the seed audit address leakage and selection risks but do not eliminate statistical overfitting or optimization variance.

TABLE III
IMPLEMENTATION DETAILS USED FOR THE MAIN EXPERIMENTS.

Component	Setting
Input tensor	10 patches × 62 channels × 50 samples
Training epochs	30
Batch size	32
Learning rate / weight decay	1×10^{-4} / 5×10^{-4}
Dropout	0.6
Hidden / encoder hidden size	128 / 512
LR scheduler	decay=0.15, gamma=0.0003
Trainable model size	Clean: 26,495,204; full residual system: 27,562,598
Trainable RA residual branch	1,067,394 parameters
Checkpoint rule	Best source-validation subject-level checkpoint
RA residual	$\alpha = 0.2$, $\rho = 0.5$, epoch=7, $\lambda = 0.10$
Metadata branch	Logistic regression; age, sex, MT, stimulation intensity
Gating policies	Validation-selected specificity/sensitivity/disagreement guards

TABLE IV
FIXED HELD-OUT CLINICAL METRICS OVER 18 TEST SUBJECTS. CONFUSION MATRIX IS TN/FP/FN/TP.

Method	Ep.F1	S.F1	Acc.	Sen.	Spe.	AUC	TN/FP/FN/TP
EA-Clean-STGNN	0.5983	0.6625	0.6667	0.7778	0.5556	0.6049	5/4/2/7
DLR-STGNN ($\lambda = 0.10$)	**0.6726**	**0.7662**	**0.7778**	**1.0000**	0.5556	**0.6543**	**5/4/0/9**
DLR-STGNN ($\lambda = 0.08$)	0.6707	**0.7662**	**0.7778**	**1.0000**	0.5556	0.6543	**5/4/0/9**

III. EXPERIMENTS AND RESULTS

A. Fixed Held-Out Test Performance

Table IV reports the fixed 18-subject held-out results. EA-Clean-STGNN achieves 0.5983 epoch F1 and 0.6625 subject F1. With the decision-level Riemannian residual, the observed epoch and subject F1 values are 0.6726 and 0.7662. The subject-level confusion matrix changes from TN/FP/FN/TP = 5/4/2/7 to 5/4/0/9: the residual corrects two false negatives without adding false positives, while specificity remains unchanged at 0.5556.

Paired resampling of the 18 subjects (20,000 bootstrap replicates) gives an observed subject-F1 difference of +0.1037 with a 95% interval of [-0.1236, 0.3290]. The interval crosses zero, so the fixed-split result is interpreted as exploratory.

B. Nested Subject-Level Robustness

Table V summarizes five-fold nested subject-level evaluation over all 89 subjects. The residual model does not improve mean nested F1 over Clean STGNN: 0.5706±0.1691 versus 0.5778±0.0617. The paired mean delta is -0.0072 with a descriptive fold-resampling interval of [-0.0966, 0.0752], and only 2/5 folds improve. Validation-selected multimodal evidence has an observed mean of 0.6230±0.1038, but its descriptive interval [-0.0190, 0.1072] crosses zero. The validation-gated metadata rule reaches 0.6384±0.0850 with a mean

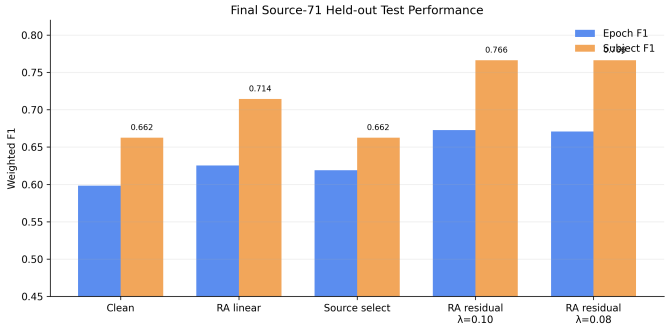


Fig. 2. Fixed held-out performance. Decision-level Riemannian residuals change the observed held-out scores; the fixed-split result is interpreted alongside the nested evaluation.

TABLE V

NESTED SUBJECT-LEVEL ROBUSTNESS OVER FIVE OUTER FOLDS. DELTA AND THE DESCRIPTIVE FOLD-RESAMPLING INTERVAL ARE PAIRED AGAINST CLEAN STGNN.

Method	Mean±Std	Delta	Desc. interval	Imp.
Clean STGNN	0.5778±0.0617	–	–	–
DLR-STGNN	0.5706±0.1691	-0.0072	[-0.0966, 0.0752]	2/5
TMS-derived fusion	0.5636±0.0477	-0.0142	[-0.0564, 0.0279]	1/5
Metadata only	0.5893±0.0645	+0.0115	[-0.0393, 0.0686]	2/5
Validation-selected multimodal	**0.6230±0.1038**	+0.0453	[-0.0190, 0.1072]	3/5
Validation-gated metadata	**0.6384±0.0850**	**+0.0607**	**[0.0091, 0.1123]**	3/5

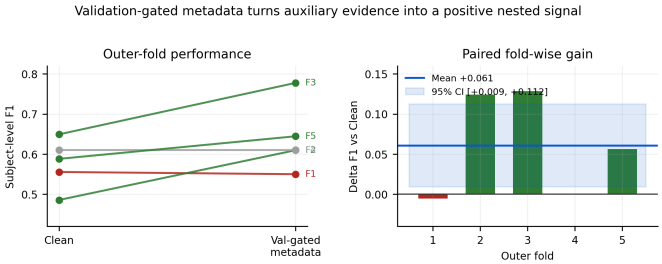


Fig. 3. Nested-fold behavior of validation-gated metadata. The rule keeps Clean STGNN as default and activates metadata only when source-validation guards support auxiliary correction. The displayed interval is a descriptive summary of five outer-fold differences.

TABLE VI

COMPRESSED NEGATIVE CONTROLS. THESE RESULTS MOTIVATE DECISION-LEVEL RATHER THAN FEATURE-LEVEL AUXILIARY EVIDENCE.

Method group	Epoch F1	Subject F1
EEGNet source-only	0.5216	0.4375
Direct Riemannian preprocessing	0.4547	0.4462
Direct log-Euclidean preprocessing	0.4760	0.4582
Late-latent N-ASN variants	0.4906–0.5274	0.3333
Early PSNA variants	0.3330	0.3333
EEGNet + N-ASN	0.3330	0.3333

delta of +0.0607 and a descriptive fold-resampling interval of [0.0091, 0.1123]. It improves 3/5 folds, leaves one unchanged because no guard is activated, and decreases one. These intervals summarize five outer-fold differences; pooled out-of-fold predictions are needed for subject-level inference across all 89 participants. Here, "TMS-derived fusion" refers to a decision-level baseline that integrates Clean STGNN logits with only raw physical stimulation parameters, namely motor threshold and stimulation intensity, while excluding general demographic metadata such as age and sex.

Fig. 3 shows the fold-wise behavior of the validation-gated rule. It improves three folds, leaves one fold unchanged by selecting no activation, and slightly reduces one fold. The fold-level improvement is therefore heterogeneous and requires pooled out-of-fold and external confirmation.

A separate five-seed audit also shows substantial optimization sensitivity. Clean STGNN and DLR-STGNN obtain subject-level F1 values of 0.5024±0.0377 and 0.5358±0.0338, respectively; the paired improvement is 0.0334±0.0462 and occurs in only two of five seeds. Nested evaluation therefore serves as the primary robustness analysis.

C. Negative Controls and Failure Analysis

Table VI compresses the main negative controls. These systematic ablations test whether stronger feature-level alignment can improve this transient TMS-evoked potential task. Direct Riemannian preprocessing is not helpful for the current waveform STGNN: Riemannian and log-Euclidean variants reduce subject F1 to 0.4462 and 0.4582. Feature-level subject-conditioned adaptation also collapses. Late-latent N-ASN variants, early PSNA variants, and EEGNet + N-ASN all reach approximately 0.3333 subject F1, indicating near single-class held-out behavior. These results characterize the present task and protocol; their generalizability to other EEG settings remains to be tested.

The EEGNet + N-ASN control extends this observation beyond STGNN. Subject-conditioned normalization also fails with a compact convolutional backbone, consistent with a mismatch between the adaptation assumption and the current cross-subject depression task. This ablation motivates further evaluation of lower-intervention decision-level evidence.

D. Subject-Level Decision Diagnostics

Because the final residual is applied after the classifier, the most relevant visualization is subject-level decision space rather than epoch-level feature space. Fig. 4 aggregates each held-out subject's epoch-level clean logits, RA logits, final probabilities, entropy, and within-subject variability. The plot provides a qualitative view of subject-specific corrections and remaining ambiguity.

IV. DISCUSSION

Direct covariance-space preprocessing was poorly matched to the current patch-wise waveform STGNN representation. EA preserves the waveform structure consumed by temporal filters and graph message passing, whereas direct Riemannian preprocessing can disturb the downstream representation.

The fixed and nested analyses of the decision-level residual differ. On the fixed held-out split, Riemannian evidence corrects two false negatives without adding false positives, increasing sensitivity and observed subject-level F1 while leaving specificity unchanged. Under nested subject-level evaluation, the F1 increase varies across partitions. The residual

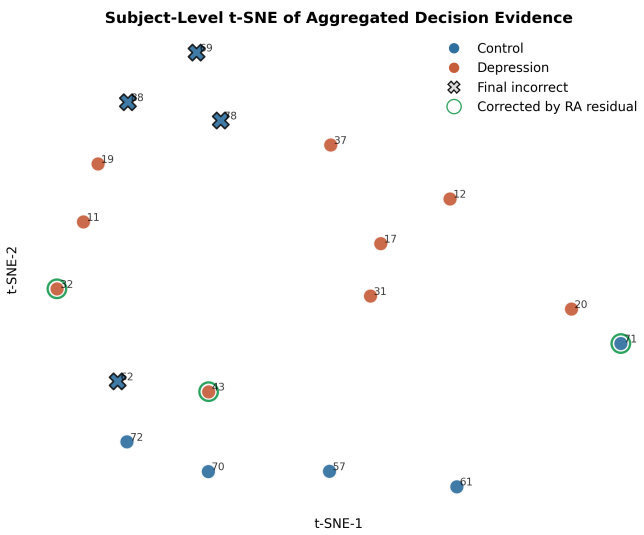


Fig. 4. Subject-level t-SNE of aggregated decision evidence. Each point is one held-out subject. Crosses mark final incorrect subjects, and green rings mark subjects corrected by the RA residual relative to the clean backbone.

branch is therefore interpreted as a partition-sensitive mechanism result.

Four non-diagnostic covariates produce a larger nested positive trend than EEG-internal auxiliary branches, and validation gating limits their activation to validation-supported cases. Clean STGNN remains the default predictor, while non-waveform metadata is used only when source-validation evidence supports an auxiliary decision rule. The clinical sufficiency and transportability of these covariates require external evaluation.

Formally, this can be viewed as

$$p_{\text{final}} = (1 - g_s)p_{\text{clean}} + g_s p_{\text{aux}}, \tag{2}$$

where g_s is selected from validation-only rules based on auxiliary validation gain, metadata confidence, and clean–auxiliary disagreement. This follows medical uncertainty-reporting principles [22] and provides a testable screening-oriented design without forcing every subject through an unstable auxiliary branch.

Clinically, the fixed held-out model is best described as a research-stage high-sensitivity screening aid. It identifies all 9 depression subjects but still produces 4 false-positive controls. This error structure may be relevant to triage, where missed cases and follow-up workload have different consequences, but the present data do not establish clinical utility. The current specificity is insufficient for standalone diagnosis, and every positive output would require standard clinical assessment. We report specificity of 0.5556 at the prespecified subject threshold of 0.5 and do not tune this threshold on the held-out test set. Although thresholds can mathematically trade sensitivity for specificity, the current 18-subject test cannot support a clinical operating-point recommendation. Probability calibration, threshold selection, false-positive workload, and decision-curve net benefit must be evaluated on independent validation and external cohorts.

A. Limitations

This study has seven main limitations. First, the fixed held-out test set contains only 18 subjects, so the observed difference has wide bootstrap uncertainty and is not a definitive split-independent effect. Second, no external cohort or multi-center validation is available. Third, specificity remains insufficient for standalone diagnosis. Fourth, the fold-resampling intervals are based on only five outer-fold differences and do not substitute for pooled out-of-fold subject-level inference. Fifth, the 26.5-million-parameter clean model is large relative to 89 independent subjects, and the five-seed audit confirms optimization sensitivity. Sixth, the current comparator set includes EEGNet but not a fully compatible recent Transformer or self-supervised EEG model. Seventh, identifying recruitment-site and record-level clinical provenance is withheld from the blinded and public materials; transparent evaluation still requires a source-verified, de-identified supplementary description of diagnostic, eligibility, treatment, missingness, and confounding variables, with ethics and consent documentation available for confidential editorial review. Future work should add pooled out-of-fold inference, lower-capacity or pretrained encoders, recent-model comparisons under identical subject splits, external calibration, decision-curve analysis, and multi-center validation.

V. CONCLUSION

We evaluated a decision-level framework for cross-subject TMS-EEG depression screening. EA-Clean-STGNN showed the most consistent performance and serves as the default waveform backbone. A Riemannian logit residual corrects two false negatives without adding false positives on the fixed split, while nested performance varies across partitions. Validation-gated non-diagnostic metadata improves three of five folds and remains an exploratory internal result. Next steps include pooled out-of-fold inference, comparisons with compatible pretrained encoders, external calibration, and multi-center validation.

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